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TCAD Based Study of 4H-SiC Charge Sheet Superjunctions with a Wide-Trench Termination

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Outline

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- Introduction: why 4H-SiC, and the PiN diode, superjunction and Charge Sheet Superjunction (CSSJ)
- Edge termination: why it is needed, the common types, and why a trench suits the CSSJ
- Literature survey: development of the CSSJ and trench terminations
- Proposed wide trench termination and simulation setup
- TCAD simulation results and comparison with reported work
- Conclusion and future work

Why 4H-SiC instead of Si?

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- A power device must block a high voltage with low leakage and conduct with a low specific on-resistance $R_{ON,sp}$; silicon devices are now close to their material limit.
- The unipolar limit [1], $R_{ON,sp} = 4V_{BR}^2 / (\mu_n \varepsilon_s E_c^3)$, shows that $R_{ON,sp}$ falls with the cube of the critical field E_c.
- 4H-SiC has $E_c \approx 2.5–3$ MV/cm, close to **10× that of Si**, which allows a thinner, more heavily doped drift layer and more than 100× lower $R_{ON,sp}$ at the same V_{BR}.
- SiC grows a native SiO_2 by thermal oxidation; the SiO_2 critical field is taken as **4 MV/cm** in this work.

Table 1: Comparison of semiconductor materials

Property	4H-SiC	Si	GaN
Bandgap (eV)	3.26	1.12	3.4
Critical field (MV/cm)	2.5–3	~0.3	~3.3
Electron mobility (cm^2/Vs)	800	1200	1000
Thermal cond. (W/cm·K)	3.8	1.5	1.3
Max. junction temp. (°C)	>600	~150	~150

Even in SiC, $R_{ON,sp}$ of a conventional drift layer still grows as V_{BR}^2. The superjunction concept removes this limitation.

[1] B. J. Baliga, Springer Handbook of Semiconductor Devices, 2022

PiN Diode, Superjunction and CSSJ

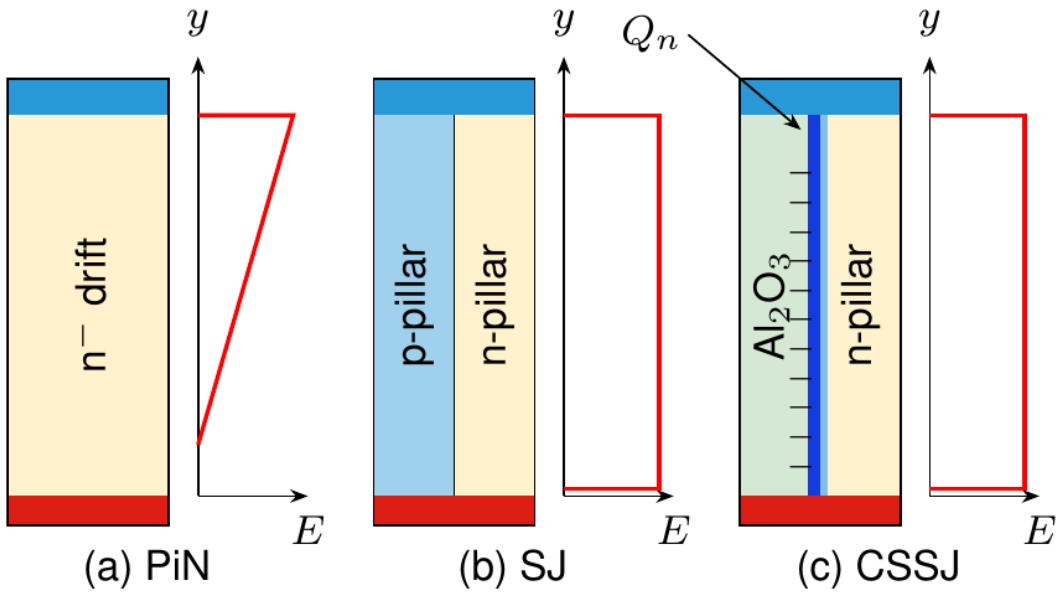


Fig. 1: Half-cells and vertical field at breakdown: (a) PiN, (b) SJ, (c) CSSJ [2]

- **PiN diode:** the field is triangular and peaks at the junction, so much of the drift layer is under-used and $R_{ON,sp} \propto V_{BR}^2$.
- **Superjunction (SJ):** charge-balanced n- and p-pillars deplete each other sideways, giving a nearly flat field and $R_{ON,sp} \propto V_{BR}$ [3]; deep p-pillars are, however, difficult to make in SiC.
- **CSSJ:** the p-pillar is replaced by a fixed negative charge Q_n = 2.5–7.9 × 10^{12} cm^{-2} at an Al_2O_3/SiO_2 interface [4, 5], which induces a hole layer that depletes the n-pillar.
- The CSSJ keeps the flat SJ field with almost the full conduction area, giving **about 40 % lower $R_{ON,sp}$** than an equivalent SJ, and needs only a trench etch and dielectric deposition [2]; the idea is used commercially in Si as SuperQ™ [6].
- This work uses Q_n = 6.27 × 10^{12} cm^{-2} (k_{eff} = 0.2), which leaves margin for process variation.

[2] K. Akshay, S. Karmalkar, IEEE J-EDS, vol. 8, 2020; [3] T. Fujihira, Jpn. J. Appl. Phys., vol. 36, 1997; [4] S. Srikanth, S. Karmalkar, Proc. IWPSD, 2007; [5] J. Buckley et al., Microelectron. Eng., vol. 80, 2005; [6] iDEAL Semiconductor, SuperQ™ press release, 2023

Edge Termination: Need and Common Types

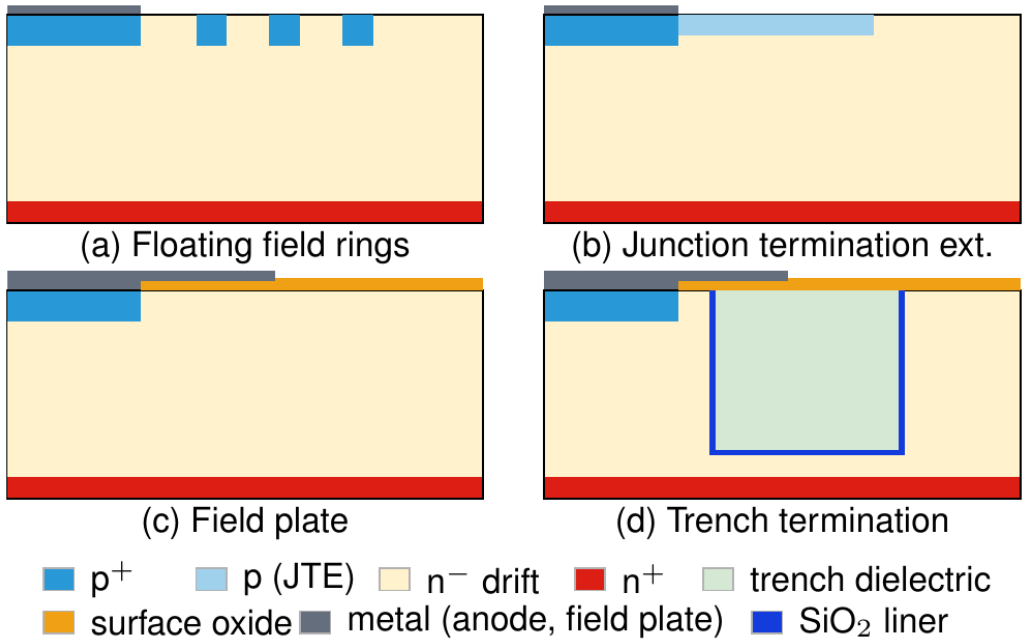


Fig. 2: Common edge termination techniques

In SJ the electric field is flat, so $V_{BR} \approx E_c \cdot D_{Drift}$. A planar termination drops V_{BR} laterally at a field $E_{lat} < E_c$, so it needs $L_{term} > D_{Drift}$, in practice several times more.

- At the device edge the last pillar has no balanced neighbour, so the potential lines crowd at the junction or trench corner and avalanche occurs well below the ideal V_{BR}; the termination efficiency is $\eta = V_{BR,term} / V_{BR,ref}$.
- **Floating field rings** [7] drop the voltage in steps across floating p-rings; they are robust but occupy a large area.
- **Junction termination extension** [8] spreads the field with a lightly doped p-region; it needs one implant, but its dose window is narrow.
- **Field plate:** the anode metal extended over an oxide pushes the potential lines away from the junction, but adds a new peak at the plate edge.
- **Bevel edge termination:** region is physically etched at edges at certain angle. Requires expert tools and additional process
- **Trench termination:** a dielectric-filled trench holds the voltage inside the dielectric, so its width stays close to the drift thickness.

[7] E. Van Brunt et al., Proc. ISPSD, 2014; [8] J. H. Zhao et al., IEEE EDL, vol. 24, 2003

Why Trench Termination?

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Table 2: Conventional vs trench termination

Feature	FLR / JTE	Trench
Termination length	~60 µm at 1200 V	~14 µm
Area efficiency	Low	High
Dose sensitivity	High (JTE)	Moderate
Breakdown efficiency	70–90 %	Up to 96 %
Suits SJ / CSSJ	Limited	Yes

Idea of present work: terminate the CSSJ with the same trench and the same charged SiO_2/Al_2O_3 stack which already exist in the active cell.

- A trench termination moves part of the voltage from the semiconductor into a **dielectric**, which can hold a much higher field without avalanche.
- Because $\varepsilon_{SiC}E_{SiC} = \varepsilon_{ox}E_{ox}$ at the interface, the normal field in SiO_2 is about 2.5× that in SiC.
- A 14 µm trench termination raised the area efficiency of a 2 A SiC diode from 62 % to about 90 % [9].
- In a CSSJ the trench and the charged dielectric **already exist** in the active cell, so no new process is needed.

[9] H. Wang et al., IEEE JESTPE, vol. 7, 2019

Literature Survey: Charge Sheet Superjunction

Table 3: Development of the CSSJ concept

Year	Ref.	Work	Contribution
2007	[4]	Srikanth & Karmalkar, IWPSD	Proposed the CSSJ: p-pillar replaced by a charge sheet
2008	[10]	Srikanth & Karmalkar, IEEE TED	Analysed the CSSJ MOSFET in Si
2020	[11]	Akshay, Jaikumar & Karmalkar, EDTM	Extended the CSSJ to 4H-SiC
2020	[2]	Akshay & Karmalkar, IEEE J-EDS	Practicability, modeling and design (1–10 kV); 5–45 % lower $R_{ON,sp}$
2022	[12]	Akshay, PhD thesis, IIT Madras	Modeling and design of superjunction devices
2023	[6]	iDEAL Semiconductor	SuperQ™: dielectric-charge SJ commercialised in Si
2025	[13]	Mohapatra et al., Semicond. Sci. Technol.	1.2 kV 4H-SiC CSSJ trench MOSFET with process design

All of these studies address the active cell only. An edge termination designed for the CSSJ has not yet been reported.

[2] K. Akshay, S. Karmalkar, IEEE J-EDS, vol. 8, 2020; [4] S. Srikanth, S. Karmalkar, Proc. IWPSD, 2007; [6] iDEAL Semiconductor, SuperQ™ press release, 2023; [10] S. Srikanth, S. Karmalkar, IEEE TED, vol. 55, 2008; [11] K. Akshay et al., Proc. IEEE EDTM, 2020; [12] K. Akshay, PhD thesis, IIT Madras, 2022; [13] R. Mohapatra et al., Semicond. Sci. Technol., vol. 40, 2025

Literature Survey: Trench Termination

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Table 4: Trench based terminations in Si, β-Ga_2O_3 and 4H-SiC

Year	Ref.	Work	Key result
2009	[14]	Deep Trench Termination (DT2): BCB-filled trench with field plate, Si p-i-n	~1300 V (1200 V class)
2018	[15]	4H-SiC: SiO_2-encapsulated spin-on dielectric trench, p-i-n	1750 V, 96 %, 14 µm
2019	[9]	4H-SiC: area-efficient trench termination	62 → ~90 % area efficiency
2021	[16]	4H-SiC: wide trench termination for SJ-SBD	Exp. 1.92 kV (TW > 20 µm)
2022	[17]	β-Ga_2O_3 SBD: trench staircase field plate	1.66 kV
2023	[18]	4H-SiC SJ: high-k TiO_2 insert in SiO_2-filled trench	Field peak moved from active corner
2024	[19]	4H-SiC SJ-SBD: ring-assisted SJ termination extension	1530 → 4150 V, 92 %

Problem statement

- None of these terminations was designed for the CSSJ; this work designs one that reuses the same trench and SiO_2/Al_2O_3 stack as the active cell.

[9] H. Wang et al., IEEE JESTPE, vol. 7, 2019; [14] L. Théolier et al., Proc. ISPSD, 2009; [15] H. Wang et al., IEEE EDL, vol. 39, 2018; [16] B. Wang et al., IEEE TED, vol. 68, 2021; [17] S. Kumar et al., Appl. Phys. Express, vol. 15, 2022; [18] Q. Zhu, Proc. IEEE COMPEL, 2023; [19] H. Cheng et al., IEEE EDL, vol. 45, 2024

Proposed Wide Trench Termination

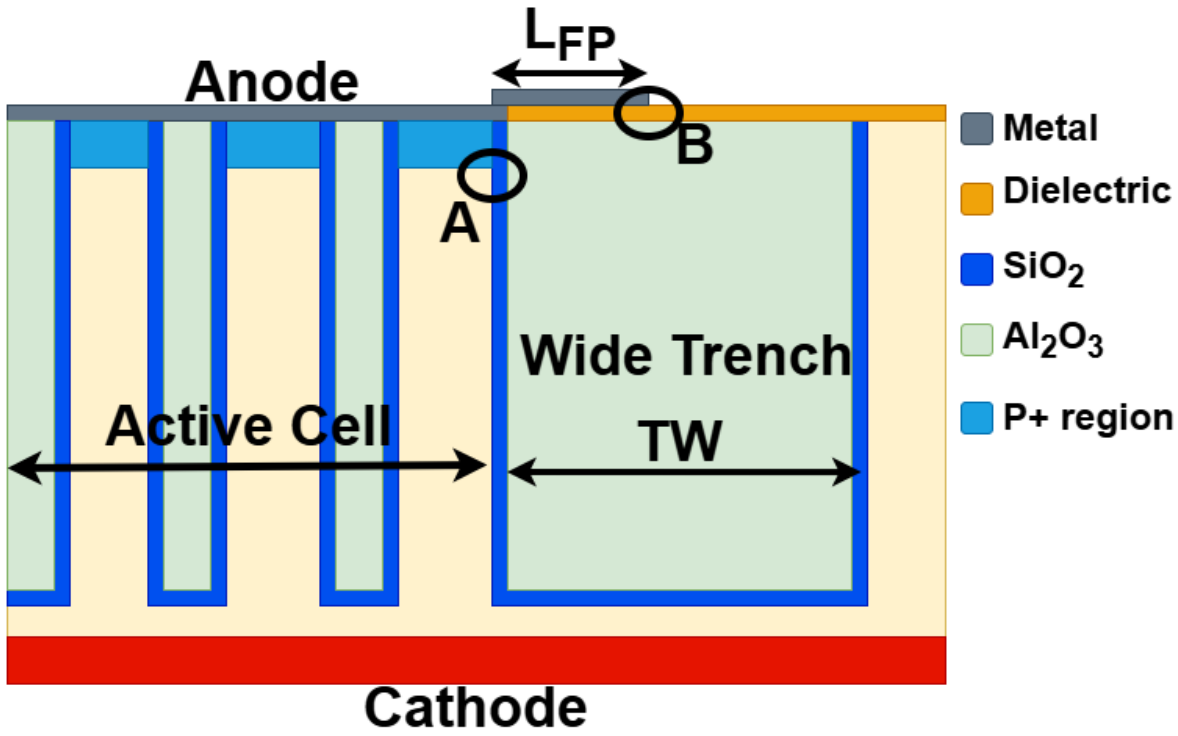


Fig. 3: Schematic cross section of the CSSJ with wide trench termination and field plate; A and B mark the field peaks

- A single **wide trench** of width TW, with the same depth and the same **SiO_2/Al_2O_3 stack** as the active trenches, is placed next to the last mesa, so Q_n is available without an extra implant.
- The anode metal extends by **L_{FP}** over a 10 nm dielectric to form a field plate that shields trench corner **A**.
- 2D Sentaurus TCAD [20] with the anisotropic Hatakeyama impact ionisation model [21].

Table 5: Main device parameters

Parameter	Value
Trench depth / drift thickness	15 µm / 16 µm
Drift doping, N_d	~5 × 10^{16} cm^{-3}
Q_n (k_{eff} = 0.2)	6.27 × 10^{12} cm^{-2}
TW / L_{FP}	1–40 µm / 4–7 µm

[20] Synopsys Sentaurus TCAD 2023.12; [21] T. Hatakeyama et al., Appl. Phys. Lett., vol. 85, 2004

Active Cell Simulation

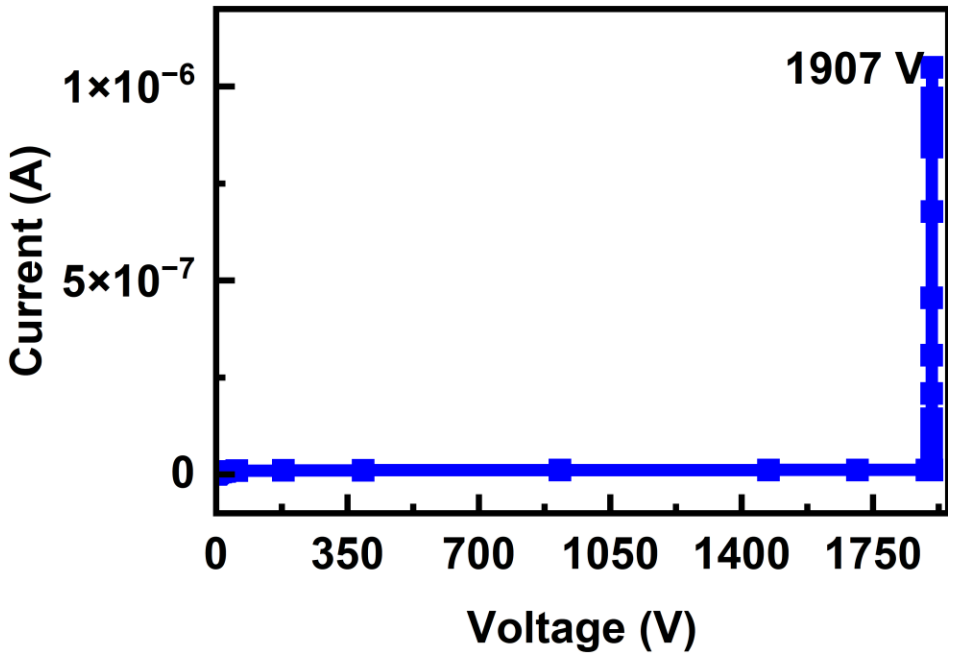


Fig. 4: Simulated reverse I–V of the active CSSJ cell

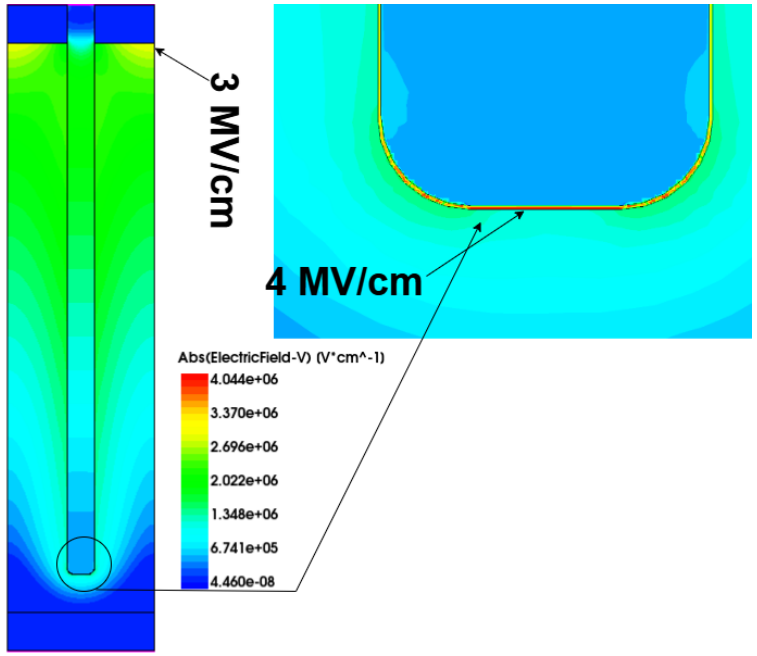


Fig. 5: Off-state electric field at breakdown

- The active cell breaks down sharply at **about 1.9 kV**; this value is used as the reference $V_{BR,ref}$ for the termination efficiency.
- The field along the mesa is nearly flat (about 3 MV/cm near the top), as expected for a superjunction, and peaks at **about 4 MV/cm** at the rounded trench bottom.

Wide Trench without Field Plate

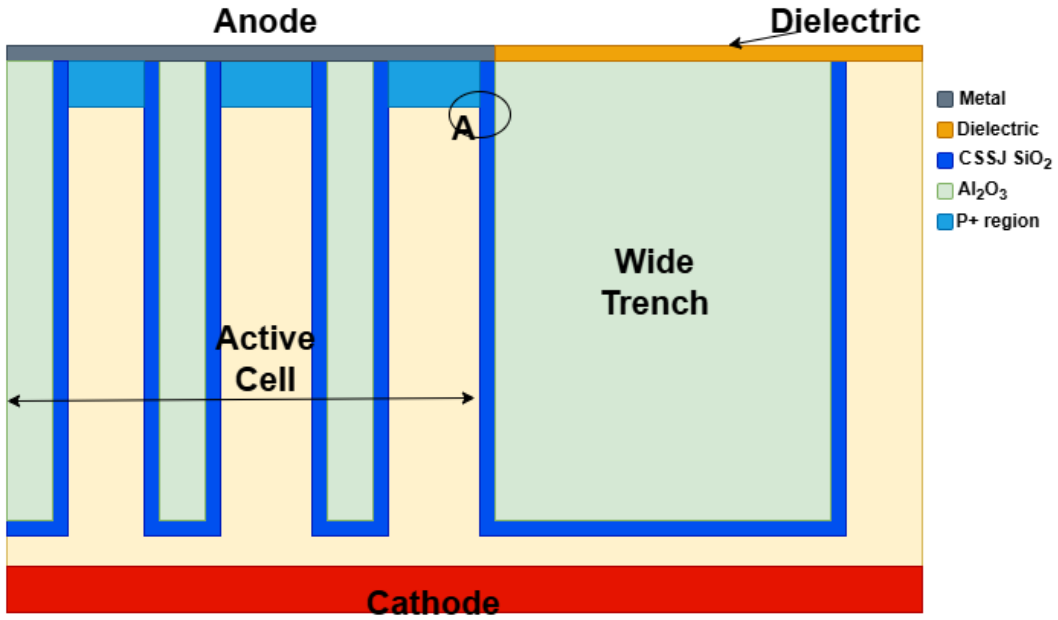


Fig. 6: Wide trench termination without field plate

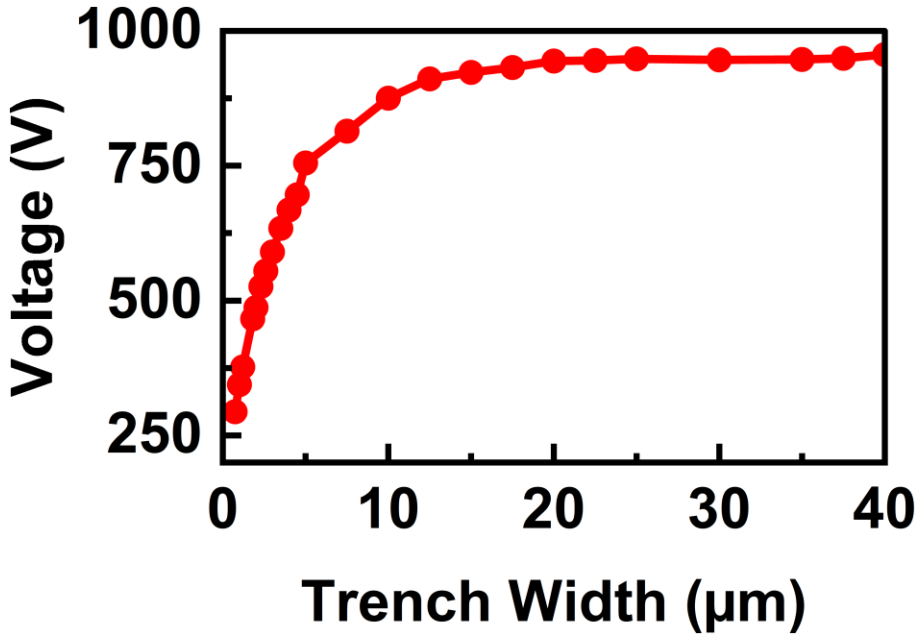


Fig. 7: Simulated V_{BR} versus trench width (L_{FP} = 0)

- With a 1 μm trench, V_{BR} is only about 300 V, because the potential cannot spread beyond the last mesa and the field crowds at the trench corner (point **A**).
- Widening the trench raises V_{BR} until it saturates at **about 940 V** (~50 % of $V_{BR,ref}$) beyond TW ≈ 20 μm; the corner field must then be reduced directly [14].

[14] L. Théolier et al., Proc. ISPSD, 2009

Field Plate and Optimised Design

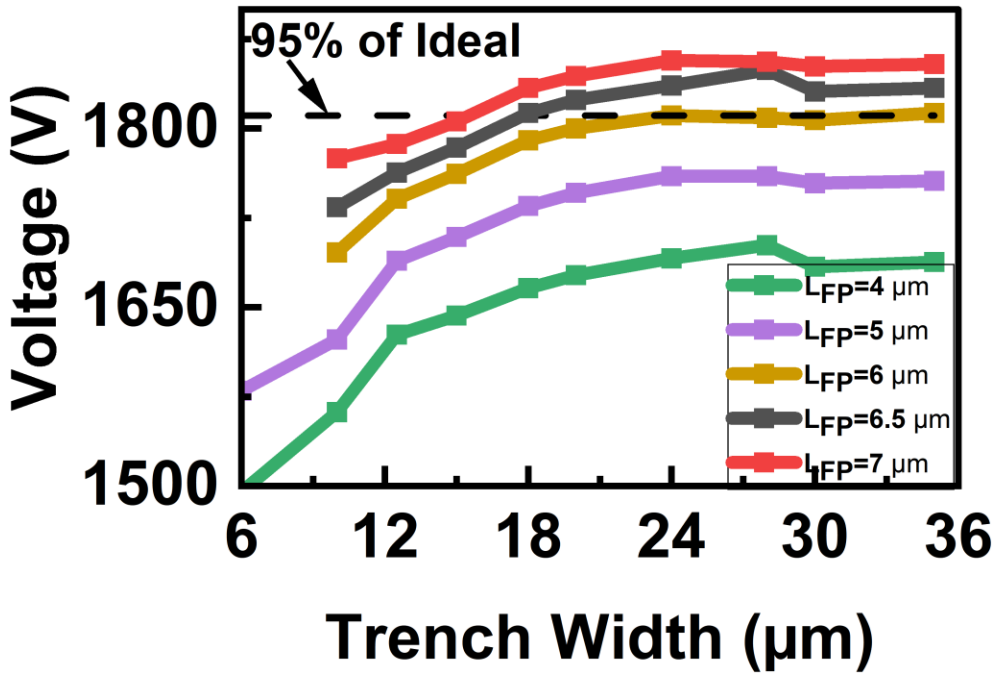


Fig. 8: Simulated V_{BR} versus trench width for different field plate lengths

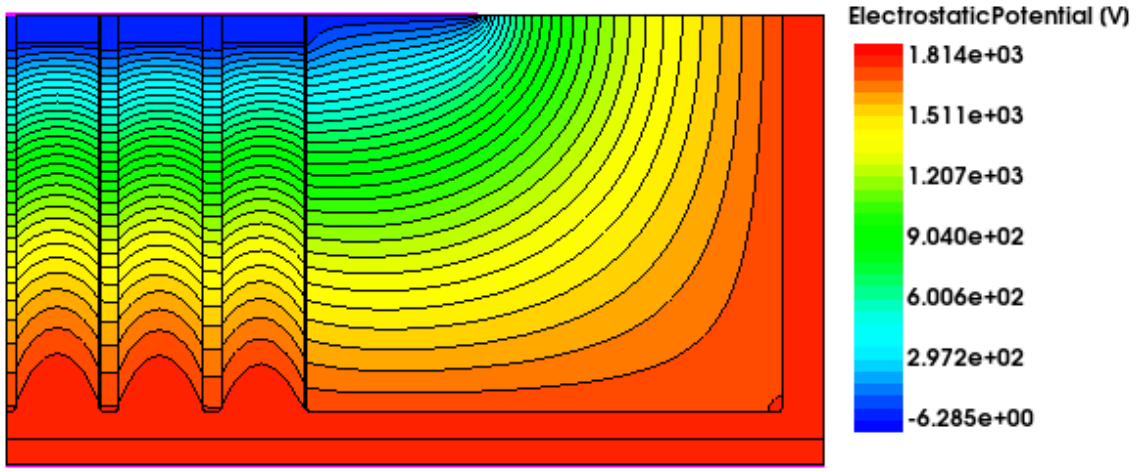


Fig. 9: Potential at breakdown for TW = 18 μm, L_{FP} = 6.5 μm

- The field plate lowers the peak at corner **A** but adds a second peak at the plate edge **B**; beyond L_{FP} = 6 μm each extra 0.5 μm adds only 20–25 V.
- With TW = **18 μm** and L_{FP} = **6.5 μm**, the termination reaches V_{BR} ≈ **1810 V (~95 % of $V_{BR,ref}$)**, and the potential lines open smoothly into the trench.

Effect of Interface Charge and Temperature

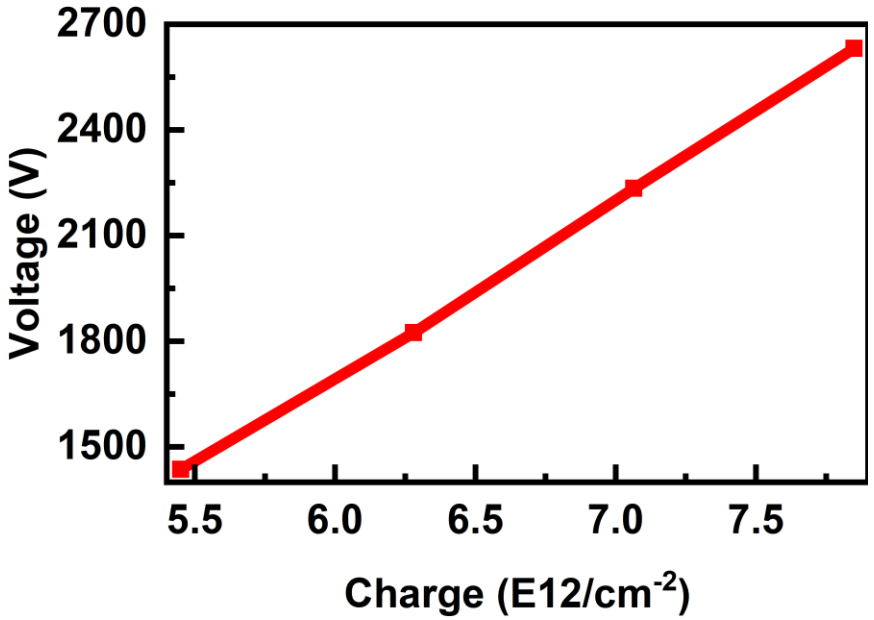


Fig. 10: V_{BR} versus negative interface charge

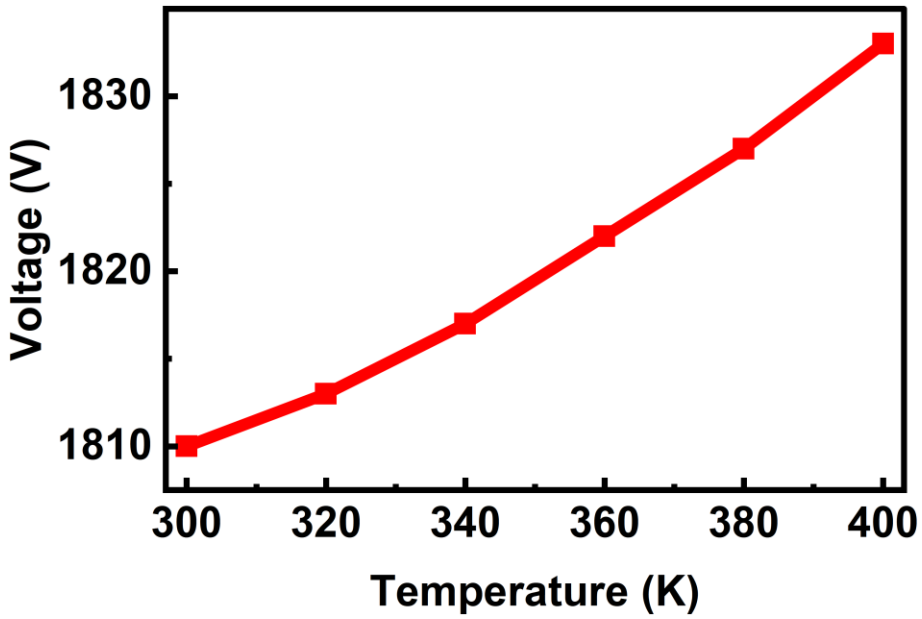


Fig. 11: V_{BR} versus temperature

- As Q_n rises from 5.45 to 7.84 × 10^{12} cm^{-2}, V_{BR} increases almost linearly from 1480 to 2630 V.
- The termination carries the same charge as the cell, so it follows the cell without re-optimisation.

- From 300 to 400 K, V_{BR} rises from 1810 to 1834 V, a coefficient of about **+0.24 V/K**.
- A positive coefficient is consistent with avalanche-dominated breakdown [22].

[22] R. Rupp et al., Proc. ISPSD, 2014

Comparison with Reported Terminations

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Table 6: Comparison of the proposed termination with trench and SJ terminations

Ref.	Device / termination	Material	V_{BR}	η	Width	Type
[14]	p-i-n, BCB-filled deep trench + FP	Si	~1300 V	–	–	Exp.
[15]	p-i-n, SiO_2-encapsulated SOD trench	4H-SiC	1750 V	96 %	14 µm	Exp.
[16]	SJ-SBD, wide trench	4H-SiC	1920 V	–	50 µm	Exp.
[17]	SBD, SiO_2-filled trench + staircase FP	β-Ga_2O_3	1660 V	–	–	Exp.
[19]	SJ-SBD, ring-assisted SJTE	4H-SiC	4150 V	92 %	–	Exp.
This work	**CSSJ, charge-sheet wide trench + FP**	**4H-SiC**	**~1810 V**	**~95 %**	**18 µm**	**Sim.**

- The proposed termination has an efficiency and width comparable to the SiO_2-encapsulated trench of [15], and is much narrower than the wide trench of [16].
- It needs no dedicated p-type termination implant or separate charge-generating stack.

[14] L. Théolier et al., Proc. ISPSD, 2009; [15] H. Wang et al., IEEE EDL, vol. 39, 2018; [16] B. Wang et al., IEEE TED, vol. 68, 2021; [17] S. Kumar et al., Appl. Phys. Express, vol. 15, 2022; [19] H. Cheng et al., IEEE EDL, vol. 45, 2024

Conclusion

- A wide trench termination has been proposed for 1700 V-class 4H-SiC CSSJs that uses the **same SiO_2/Al_2O_3 stack and interface charge** as the active cell.
- The active and termination trenches can be etched with one mask and receive the same liner and ALD Al_2O_3 [2, 13], so no extra implant or process step is needed.
- Without a field plate, the termination saturates at about 940 V, limited by the field at the trench corner.
- With a field plate on 10 nm dielectric, it reaches **~1810 V (~95 % of $V_{BR,ref}$)** at TW = 18 µm and L_{FP} = 6.5 µm.
- The termination tracks the active cell when Q_n varies, and its +0.24 V/K coefficient indicates avalanche breakdown.

Table 7: Summary of termination results

	Without FP	With FP
TW	~20 µm	18 µm
V_{BR}	~940 V	~1810 V
η	~50 %	~95 %

Future work

- Examine the dielectric field under the field plate edge.
- Fabricate and verify the design experimentally.

[2] K. Akshay, S. Karmalkar, IEEE J-EDS, vol. 8, 2020; [13] R. Mohapatra et al., Semicond. Sci. Technol., vol. 40, 2025

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Thank You

Questions?

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